Sai Krishna Nerella | +91 9176762559 | nsaikrishna94@gmail.com

PROFESSIONAL SUMMARY

Senior Software Engineer with 10 years of experience building high-scale backend and platform systems in fintech, logistics, and healthcare. Now focused on AI Platforms, that includes model inferencing, feature engineering, MLOps and DevOps.

WORK EXPERIENCE

Senior Software Engineer (MTS 1), PayPal India Private Limited Feb 2021 – Present

AI Platform – Model Inferencing & Feature Engineering May 2025 – Present

Technologies: Kubernetes, GCP, Harness, Jenkins, Argo CD, Java, Spring Boot, MySQL, Docker, Prometheus

- **Model Inferencing Platform Cluster Setup:** Architected and stood up new on-prem and cloud clusters to replace the legacy platform, owning the storage, network, and monitoring layers in partnership with multiple infrastructure teams. Serving **10M requests/day** today, with capacity designed for **500M requests/day** at full ramp.
- **GPU Optimization:** Prototyped GPU-aware autoscaling for real-time workloads using KEDA, HPA, and DCGM metrics, reclaiming idle GPUs, with projected savings of **$400K/year** once rolled out to production.
- **CI/CD Maturity:** Owned the org's migration from Jenkins and Argo to Harness (company-wide CI/CD initiative), delivering reference pipelines for Java, Go, and Python that became templates for **15 more components**.

DropZone – Secure File Exchange Platform May 2023 – Apr 2025

Technologies: Java, Scala, MySQL, SFTP, ReactJS, Elasticsearch, Spring Boot

- **HTTPS Connector:** Designed a reusable HTTPS connector for file uploads and downloads, giving multiple business teams a single, standard way to exchange files with external vendors.
- **Self-Servicing Capabilities:** Added a DropZone portal feature that lets clients view vendor server file metadata themselves, removing manual access and cutting support tickets by about **50 per month**.

Location Services Platform Feb 2021 – Apr 2023

Technologies: Java, Python, BigQuery, Kafka, Spring Boot

- **Vendor-Agnostic Address Normalization System:** Built an address normalization and suggestion service integrated with **3 address vendors**, replacing a legacy system and handling **120M requests/day**. Strengthened security standards and streamlined client onboarding.
- **Application-Context-Based Authorization:** Inner-sourced an annotation-driven authorization feature into the shared API platform, letting teams control access to internal APIs by calling application.
- **Offline Address Batch Processing:** Led a **team of two** to build a batch pipeline that cleans and standardizes historical address data for multiple internal teams, handling datasets of **100M–600M addresses**.
- **Self-Sustenance Mode for Address Management:** Made the Location Services Platform largely self-running by putting in place SOPs, metrics, alerts, and automation scripts, minimizing manual intervention.

Software Engineer II, Blackbuck Limited Jun 2019 – Dec 2020

Technologies: Java, MySQL, PostgreSQL, Elasticsearch, Presto, New Relic, Python, Spring Boot, Kafka, RabbitMQ

- **Payment Service:** Built a microservice processing **100K daily payments** to truck owners across each stage of shipment, replacing a monolithic system.
- **Wallet Service:** Helped build a wallet system for Blackbuck suppliers, supporting **50K daily transactions** for fuel and truck-related purchases.
- **Cost and Time Savings in Income Report Calculation Systems:** Saved **$300K/year** in AWS costs by migrating **1B records** from Redshift to PostgreSQL with no loss in performance, and cut profitability report generation from **2–3 hours to 5–10 minutes** using thread pools and indexed search columns.

Software Engineer, Medlife International Pvt Ltd, Bangalore Jul 2016 – May 2019

Technologies: ReactJS, Java, Spring Boot, MongoDB, AWS, Node.js, Nginx

- **Discount Orchestration Service:** Built an orchestration service that computes the net discount by aggregating product, coupon, and wallet discounts from downstream services, handling **500K requests/day**.
- **Medlife Portals:** Built full-stack portals for Medlife Labs (**50K requests/day**), Medlife Products (**1K/day**), and Medlife Franchise (**1K/day**).

EDUCATION

- Indian Institute of Technology Madras – B.Tech in Aerospace Engineering 2012 – 2016
- Bhashyam Academy 2010 – 2012

TECHNICAL SKILLS

- **Languages:** Java, Scala, Python, Go, JavaScript
- **Frameworks & Libraries:** Spring Boot, ReactJS, Node.js, TestNG, JUnit
- **Cloud & Infrastructure:** Kubernetes, Docker, GCP, AWS, Nginx
- **CI/CD & Tooling:** Harness, Jenkins, Argo CD, Git
- **Autoscaling & Observability:** KEDA, HPA, DCGM, New Relic, Prometheus
- **Data Stores & Messaging:** MySQL, PostgreSQL, MongoDB, Elasticsearch, BigQuery, Redshift, Presto, Kafka, RabbitMQ

SCHOLASTIC ACHIEVEMENTS

- Secured All India Rank 1856 (GE) in IIT-JEE and State Rank 563 (GE) in AIEEE